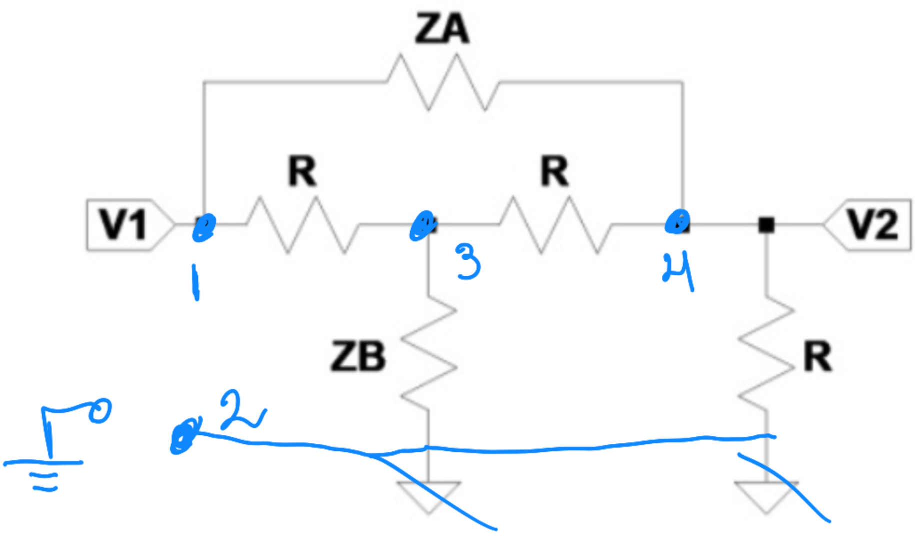
Ejercicio #5

Dado el siguiente cuadripolo cargado con R y cumpliendo con la condición de diseño $Z_A . Z_B = R^2$ La red se conoce como ***Red de Resistencia constante***.

Aplique MAI para obtener $\frac{V_2}{V_1}$ y Z_{in}



$$Y_{MAI} = \begin{pmatrix} Y_A+G & 0 & -G & -Y_A \\ 0 & Y_B+G & -Y_B & -G \\ -G & -Y_B & Y_B+2G & -G \\ -Y_A & -G & -G & Y_A+2G \end{pmatrix}$$

$\rightarrow \sum$ columnas $= 0$

$\rightarrow \sum$ filas $= 0$

$\Big\}$ concuerda con los criterios de la MAI

Procedimiento de Armado

↳ se utiliza un nodo de referencia externo ∴ GND es un NODO. En este caso el nodo 2.

↳ La fila (i) es el nodo y las columnas (j) es la relación del nodo iésimo con el jésimo

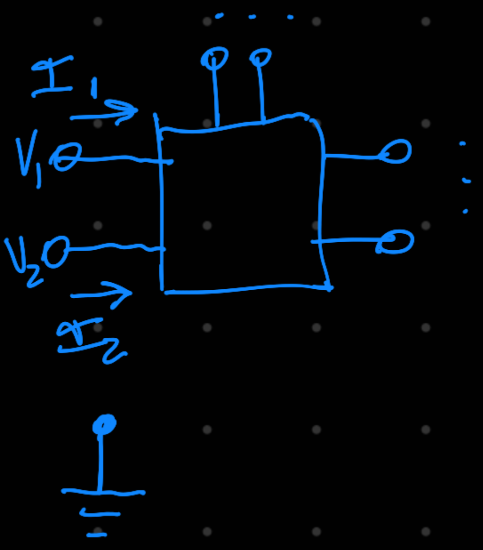
$$\begin{pmatrix} y_{11} & y_{12} \cdots y_{1n} \\ y_{21} & \ddots & \vdots \\ \vdots & & \ddots \end{pmatrix} \begin{array}{l} \leftarrow \text{nodo } 1 \\ \leftarrow \text{nodo } 2 \\ \leftarrow \text{nodo } n \end{array}$$

↳ Los valores y_{ii} (diagonal principal) son la suma de las admitancias conectadas al nodo.

↳ Los valores y_{ij} es la admitancia (con signo menos) entre nodo i y nodo j

* Solo conexión directa, es decir, sin pasar por otro componente o nodo.

Propiedades y Definición



$$\vec{I} = \vec{Y}_{MAI}\,\vec{V} + \cancel{\vec{I}_0}$$

0 gralmente

fuentes internas

→ Relación de todas las corrientes y todas las tensiones

→ $Y_{ij} = \dfrac{I_i}{V_j}\Big|_{V_k=0,\ \forall k=j}$

superposición
↳ se "pasivan" los demás nodos que no sean j

→ Y_{MAI} es simétrica de $N \times N$ siendo N la cantidad de nodos

→ Σ filas $= 0$ & Σ columnas $= 0$

(no hay pérdidas) (malla cerrada)

Funciones de RED

↳ Impedancia de un Puerto: Z_{ij}

↳ Transimpedancia: $Z_{ij\,mn}$

↳ Transferencia de tensión: $A_{ij\,mn}$

→ Cofactor Primer Orden

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Determinante matriz menor

M_{ij} es el determinante de la matriz resultante de eliminar la fila i y columna j

* Como Y_{MAI} es simétrica

$$C_{ij} = C_{ji}$$

→ Cofactor Segundo orden

$$C_{(i_1,i_2)(j_1,j_2)} = (-1)^{(i_1+i_2)+(j_1+j_2)} \det\left(A_{(i_1,i_2),(j_1,j_2)}\right)$$

filas (i_1, i_2) y columnas (j_1, j_2)

→ transimpedancia

$$Z_{mn}^{ij} = \frac{V_{ij}}{I_{mn}} = \frac{C_{ij\,mn}}{C_{mn}} = \frac{C^2}{C_1}$$

← se puede usar C_{nn}

→ Impedancia

$$Z_{mn} = \frac{V_{mn}}{I_{mn}} = \frac{C_{mn\,mn}}{C_{mn}}$$

→ transferencia de tensión

$$A_{mn}^{ij} = \frac{V_{ij}}{V_{mn}} = \frac{V_{ij}}{I_{mn}} \cdot \frac{1}{\frac{V_{mn}}{I_{mn}}} \leftarrow \frac{1}{Z_{mn}}$$

$$A_{mn}^{ij} = \frac{C_{ij\,mn}}{C_{mn\,mn}}$$

→ Continuando con el ejercicio

$$A^{42}_{12} = \frac{\text{etiqueta } V_2}{\text{etiqueta } V_1}$$

$$\begin{pmatrix} Y_A+G & 0 & -G & -Y_A \\ 0 & Y_B+G & -Y_B & -G \\ -G & -Y_B & Y_B+2G & -G \\ -Y_A & -G & -G & Y_A+2G \end{pmatrix}$$

$$A^{42}_{12} = \frac{C_{(42)(12)}}{C_{(12)(12)}} = \frac{(-1)^9 \begin{vmatrix} -G & -Y_A \\ Y_B+2G & -G \end{vmatrix}}{(-1)^6 \begin{vmatrix} Y_B+2G & -G \\ -G & Y_A+2G \end{vmatrix}}$$

$$A = \frac{-1\left[G^2 + Y_A Y_B + 2G Y_A\right]}{Y_B Y_A + Y_B 2G + Y_A 2G + 3G^2}$$

→ si $Z_A \cdot Z_B = R^2 \rightarrow \frac{1}{Z_A} \cdot \frac{1}{Z_B} = \frac{1}{R^2} \Rightarrow Y_A Y_B = G^2$

$$A = \frac{-1\left[G^2 + Y_A Y_B + 2G Y_A\right]}{Y_B Y_A + Y_B 2G + Y_A 2G + 3G^2} = \frac{-(2G^2 + 2G Y_A)}{4G^2 + 2G(Y_A + Y_B)}$$

$$A = -1\,\frac{G + Y_A}{2G + (Y_A + Y_B)}$$

$$Z_{in} = Z_{12} = \frac{C_{12\,12}}{C_{12}} = \frac{G^2 + \underbrace{(Y_B + Y_A)2G}_{Y_B Y_A + Y_B 2G + Y_A 2G} + 3G^2}{?}$$

$$\begin{pmatrix} \cancel{Y_A+G} & \cancel{0} & \cancel{-G} & \cancel{-Y_A} \\ 0 & \cancel{Y_B+G} & -Y_B & -G \\ -G & \cancel{-Y_B} & Y_B+2G & -G \\ -Y_A & \cancel{-G} & -G & Y_A+2G \end{pmatrix} \rightarrow \begin{pmatrix} 0 & -Y_B & -G \\ -G & Y_B+2G & -G \\ -Y_A & -G & Y_A+2G \end{pmatrix}$$

$$\rightarrow$$

$$C_{12} = (-1)^{1+2}\left[Y_B\left(-G(Y_A+2G) - G Y_A\right) + -G\left(G^2 + Y_A(Y_B+2G)\right)\right]$$

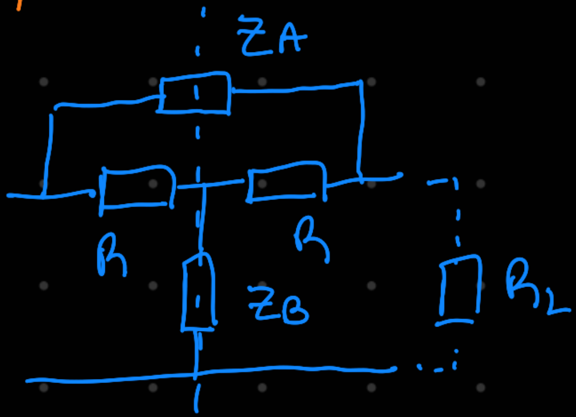
$$= -1\left[-2G Y_A Y_B - 2G^2 Y_B - G^3 - G Y_A Y_B - 2G^2 Y_A\right]$$

$$= G^3 + 2G^2(Y_A + Y_B) + 3G^3$$

$$\therefore Z_{in} = \frac{4G^2 + 2G(Y_A+Y_B)}{4G^3 + 2G^2(Y_A+Y_B)} = \frac{\cancel{[2G^2 + G(Y_A+Y_B)]}}{G\,\cancel{[2G^2 + G(Y_A+Y_B)]}}$$

$$Z_{in} = \frac{1}{G} = R$$

Cuadripolos



$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases}$$

es simétrico $\to$ no considero R_L conectado

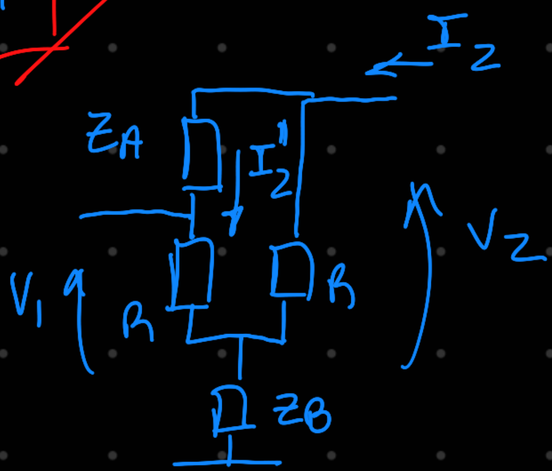
$Z_{11} = Z_{22}$

$$\to \left.\frac{V_1}{I_1}\right|_{I_2=0} = Z_{11} \to \left[R // (Z_A + R)\right] + Z_B$$

$$\to \frac{R(Z_A+R)}{2R+Z_A} + Z_B$$

$$\to \left.\frac{V_2}{I_2}\right|_{I_1=0} = Z_{22} \to Z_{11}$$

$$\to \left.\frac{V_1}{I_2}\right|_{I_1=0} = Z_{12} \to$$



$$V_1 = R \cdot I_2'' + Z_B I_2 \quad // \quad I_2'' = I_2 \frac{R}{Z_A + 2R}$$

$$V_1 = I_2 \left(Z_B + \frac{R^2}{Z_A + 2R} \right)$$

$$Z_{12} = Z_B + \frac{R^2}{Z_A + 2R} = Z_{21}$$

$$Z = \begin{pmatrix} \frac{R(Z_A + R)}{2R + Z_A} + Z_B & Z_B + \frac{R^2}{Z_A + 2R} \\ Z_B + \frac{R^2}{Z_A + 2R} & \frac{R(Z_A + R)}{2R + Z_A} + Z_B \end{pmatrix}$$

$\rightarrow$ Conversión $Z \rightarrow T$

$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases} \quad // \quad \begin{cases} V_1 = A V_2 + B(-I_2) \\ I_1 = C V_2 + D(-I_2) \end{cases}$$

$$A = \left. \frac{V_1}{V_2} \right|_{-I_2 = 0} = \frac{Z_{11}}{Z_{21}}$$

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0} \Rightarrow V_1 = z_{11} I_1 + z_{12} I_2$$

$$-z_{22} I_2 = z_{21} I_1$$

$$-I_2 \frac{z_{22}}{z_{21}} = I_1$$

$$V_1 = z_{11} \, -I_2 \left(\frac{z_{22}}{z_{21}}\right) + z_{12} I_2$$

$$\frac{V_1}{-I_2} = \frac{z_{11} z_{22} - z_{12} z_{21}}{z_{21}} = \frac{\Delta z}{z_{21}}$$

$$C = \left.\frac{I_1}{V_2}\right|_{-I_2=0} \Rightarrow \frac{I_1}{V_2} = \frac{1}{z_{21}}$$

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0} \Rightarrow -z_{22} I_2 = z_{21} I_1$$

$$\frac{I_1}{-I_2} = \frac{z_{22}}{z_{21}}$$

$$Z \to T : \frac{1}{z_{21}} \begin{pmatrix} z_{11} & \Delta z \\ 1 & z_{22} \end{pmatrix}$$

$$\Delta z = \left[\frac{R(z_A+R)}{2R+z_A} + z_B\right]^2 - \left[z_B + \frac{R^2}{z_A+2R}\right]^2$$

$$\left[\frac{R^2(z_A+R)^2}{(2R+z_A)^2} + \frac{2z_B R(z_A+\cancel{R})}{2R+z_A} + \cancel{z_B^2}\right]$$

$$-$$

$$\left[\cancel{z_B^2} + \cancel{\frac{2z_B R^2}{z_A+2R}} + \frac{R^4}{(z_A+2R)^2}\right]$$

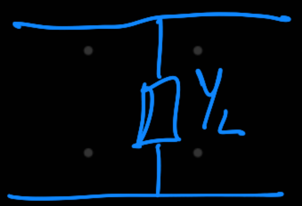
$$\frac{R^2(z_A+R)^2}{(2R+z_A)^2} + \frac{2R^2(2R+z_A)}{(2R+z_A)^2} - \frac{R^4}{(2R+z_A)^2}$$

$$\frac{R^2\left[(z_A+R)^2 + 2(2R+z_A) - R^2\right]}{(2R+z_A)^2}$$

$$\Delta Z = \frac{R^2\left[Z_A^2 + 2Z_A R + \cancel{R^2} + 4R + 2Z_A - \cancel{R^2}\right]}{4R^2 + 4R Z_A + Z_A^2}$$

$$T = \frac{Z_A + 2R}{2 Z_B R + 2R^2}\begin{pmatrix} \frac{R(Z_A+R)}{2R+Z_A} + Z_B & \Delta Z \\ 1 & \frac{R(Z_A+R)}{2R+Z_A} + Z_B \end{pmatrix}$$

$T_{R_2} \longrightarrow$



$\longrightarrow \begin{pmatrix} 1 & 0 \\ Y_L & 1 \end{pmatrix}$

$$\begin{cases} V_1 = A V_2 + B(-I_2) \\ I_1 = C V_2 + D(-I_2) \end{cases}$$

$$A = \left.\frac{V_1}{V_2}\right|_{-I_2=0} = 1$$

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0} = \quad V_1 = V_2 \quad \frac{V_2}{I_2} = Z_{in} = 0$$

$$C = \frac{I_1}{V_2}\Bigg|_{-I_2=0} = \quad V_1 = V_2 \quad \frac{I_1}{V_1} = Y_L$$

$$D = \frac{I_1}{-I_2}\Bigg|_{V_2=0} \rightarrow I_1 = -I_2 \rightarrow 1$$

$$T_{total} = T \cdot T_L = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \quad // \quad A = \frac{V_1}{V_2}$$

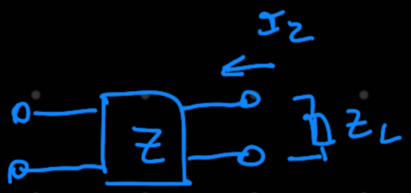
$$\frac{Z_A + 2R}{2 Z_B R + 2R^2} \begin{pmatrix} \frac{R(Z_A+R)}{2R+Z_A} + Z_B & \Delta Z \\ 1 & \frac{R(Z_A+R)}{2R+Z_A} + Z_B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ Y_L & 1 \end{pmatrix}$$

$$A = \frac{Z_A + 2R}{2 Z_B R + 2R^2} \left(\frac{R(Z_A+R)}{2R+Z_A} + Z_B + \Delta Z\, Y_L \right)$$

Y $\frac{V_2}{V_1} = \frac{1}{A}$ (la transferencia es igual a la inversa del parámetro A)

→ se verifica de manera computacional

Cuadripolo Z cargado



$$-I_2 Z_L = V_2$$

$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases}$$

$$Z_{in} = \frac{V_1}{I_1} \Rightarrow \frac{V_1}{I_1} = Z_{11} + Z_{12}\frac{I_2}{I_1}$$

$$\frac{V_2}{I_1} = Z_{21} + Z_{22}\frac{I_2}{I_1} \longrightarrow \frac{-I_2 Z_L}{I_1} - Z_{22}\frac{I_2}{I_1} = Z_{21}$$

$$-\frac{I_2}{I_1}(Z_L + Z_{22}) = Z_{21} \Rightarrow \frac{-Z_{21}}{Z_L + Z_{22}} = \frac{I_2}{I_1}$$

$$Z_{in} = Z_{11} + Z_{12}\frac{-Z_{21}}{Z_L + Z_{22}} = Z_{11} - \frac{Z_{12} Z_{21}}{Z_L + Z_{22}}$$

$$\begin{pmatrix} \frac{R(Z_A+R)}{2R+Z_A} + Z_B & Z_B + \frac{R^2}{Z_A+2R} \\ Z_B + \frac{R^2}{Z_A+2R} & \frac{R(Z_A+R)}{2R+Z_A} + Z_B \end{pmatrix}$$

de los Parametros Z

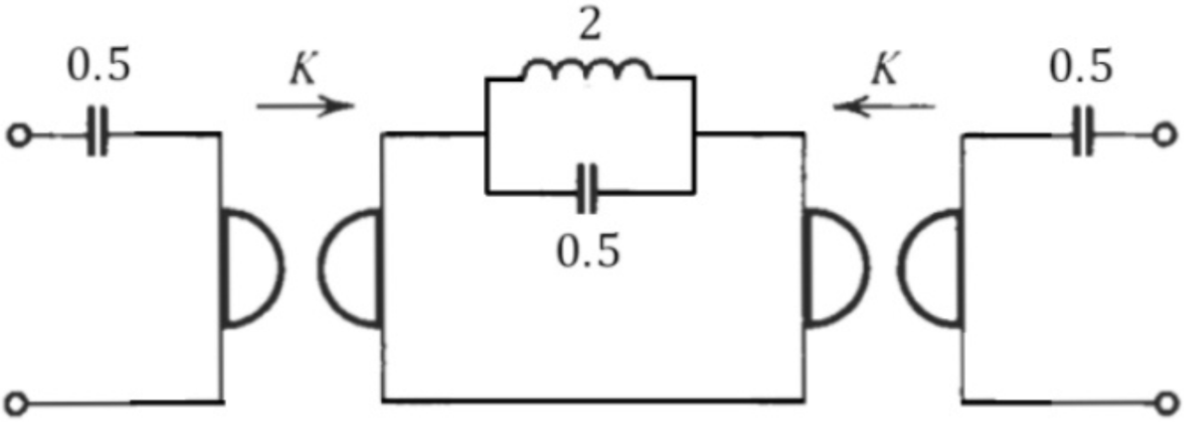
Zin con Z_L

$$Z_{in} = \frac{R(Z_A+R)}{2R+Z_A} + Z_B - \frac{\left(Z_B + \frac{R^2}{Z_A+2R}\right)^2}{\frac{R(Z_A+R)}{2R+Z_A} + Z_B + Z_L}$$

→ Se verifica de manera computacional → debería ser $Z_{in} = R = \frac{1}{G}$

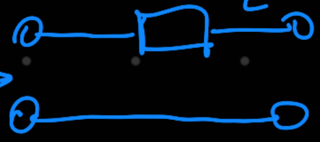
Ejercicio #6

Obtenga un circuito lattice equivalente al siguiente cuadripolo. Utilice la constante del girador $K = \sqrt{2}$



$K = \beta^2 = \sqrt{2} \rightarrow \beta = 2$

$T_{Z_L} \rightarrow$



$\rightarrow \begin{pmatrix} 1 & Z_L \\ 0 & 1 \end{pmatrix}$

$$\begin{cases} V_1 = A V_2 + B(-I_2) \\ I_1 = C V_2 + D(-I_2) \end{cases}$$

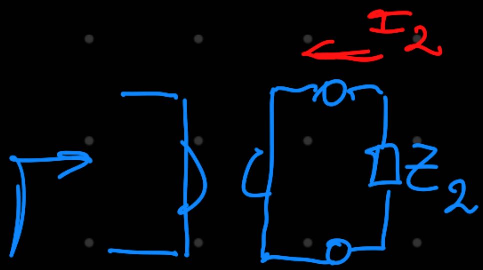
$$A = \left.\frac{V_1}{V_2}\right|_{-I_2=0} = 1$$

$$B = \left.\frac{V_1}{-I_2}\right|_{V_2=0} = Z_L$$

$$C = \left.\frac{I_1}{V_2}\right|_{-I_2=0} = I_1 = 0 = 0$$

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0} \rightarrow I_1 = -I_2 = 1$$

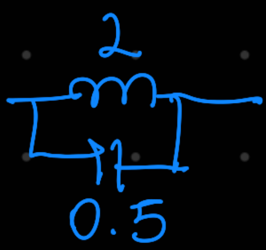
$$T_{girador} = \begin{pmatrix} 0 & -R \\ -\frac{1}{R} & 0 \end{pmatrix}$$



$$Z_{in} = \frac{R^2}{Z_2}$$

$$\left.\begin{array}{l} \rightarrow V_1 = R I_2 \\ \rightarrow I_1 = -\frac{V_2}{R} \end{array}\right) \%$$

$$\frac{V_1}{I_1} = \frac{I_2}{V_2} \cdot -R^2 \longrightarrow Z_1 = \frac{R^2}{Z_2}$$

$Z \Rightarrow$ 

$$\rightarrow \left(\frac{1}{2s} + \frac{s}{2}\right)^{-1} = \left(\frac{1+s^2}{2s}\right)^{-1} = \frac{2s}{s^2+1}$$

$$T_{total} = \begin{pmatrix} 1 & \frac{2}{s} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -2 \\ -\frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & \frac{2s}{s^2+1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -2 \\ -\frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & \frac{2}{s} \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -\frac{1}{s} & -2 \\ -\frac{1}{2} & 0 \end{pmatrix}$$

$$\frac{-2-2(s^2+1)}{s^2+1} = -\frac{2s^2+4}{s^2+1}$$

$$\begin{pmatrix} -\frac{1}{s} & -\frac{2}{s^2+1} - 2 \\ -\frac{1}{2} & -\frac{s}{s^2+1} \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} -\frac{1}{s} & -\frac{2s^2+4}{s^2+1} \\ -\frac{1}{2} & -\frac{s}{s^2+1} \end{pmatrix} \begin{pmatrix} 0 & -2 \\ -\frac{1}{2} & 0 \end{pmatrix}} \begin{pmatrix} 1 & \frac{2}{s} \\ 0 & 1 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} \frac{2s^2+4}{2s^2+2} & \frac{2}{s} \\ \frac{s}{2s^2+2} & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{2}{s} \\ 0 & 1 \end{pmatrix}}$$

$$T_{total} = \begin{pmatrix} \frac{s^2+2}{s^2+1} & \frac{4s^2+6}{s(s^2+1)} \\ \frac{1}{2\frac{s^2+1}{s}} & \frac{s^2+2}{s^2+1} \end{pmatrix}$$

$$\frac{\frac{s^2+2}{s^2+1}}{} \cdot \frac{\frac{2}{s}}{\frac{2}{s}} = \frac{\frac{s^2+2}{s}\,2}{\frac{s^2+1}{s}\,2}$$

$z_{11} = z_{22}$

z_{21}

$$\rightarrow\ Z \rightarrow T : \frac{1}{z_{21}} \begin{pmatrix} z_{11} & \Delta z \\ 1 & z_{22} \end{pmatrix}$$

$$Z = \begin{pmatrix} \frac{2s^2+4}{s} & ? \\ \frac{2s^2+2}{s} & \frac{2s^2+4}{s} \end{pmatrix}$$

Salen por inspección visual menos z_{12}

$$\begin{cases} V_1 = A\, V_2 + B\, -I_2 \\ I_1 = C\, V_2 + D\, -I_2 \end{cases}$$

$V_1 = z_{11} I_1 + z_{12} I_2$

$V_2 = z_{21} I_1 + z_{22} I_2$

$D I_2 = C V_2$

$V_2 = I_2 \frac{D}{C}$

$$z_{12} = \frac{V_1}{I_2}\bigg|_{I_1=0}$$

$$V_1 = A\, I_2 \frac{D}{C} + B\,(-I_2)$$

$$V_1 = I_2 \left(\frac{AD - BC}{C} \right)$$

$$\frac{V_1}{I_2} = \frac{\Delta T}{C} = Z_{12}$$

$$\Delta T = \left(\frac{s^2+2}{s^2+1}\right)^2 - \frac{4s^2+6}{\cancel{s}(s^2+1)} \; \frac{\cancel{s}}{(s^2+1)\,2}$$

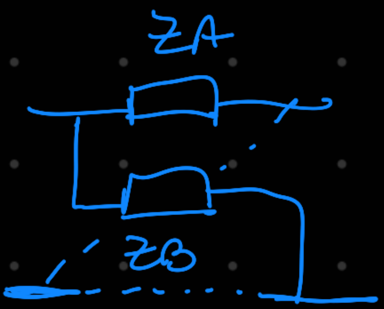
$$\frac{(s^2+2)^2}{(s^2+1)^2} - \frac{2s^2+3}{(s^2+1)^2} = \frac{s^4+4s^2+4-2s^2-3}{(s^2+1)^2}$$

$$\frac{s^4+2s^2+1}{(s^2+1)^2} = \frac{(s^2+1)^2}{(s^2+1)^2} = 1$$

$$\therefore\; z_{12} = \frac{1}{C} = z_{21}$$

$$Z = \begin{pmatrix} \frac{2s^2+4}{s} & \frac{2s^2+2}{s} \\ \frac{2s^2+2}{s} & \frac{2s^2+4}{s} \end{pmatrix}$$

y la matriz Z de un lattice simétrico es:



$$\Rightarrow \begin{pmatrix} \frac{Z_B+Z_A}{2} & \frac{Z_B-Z_A}{2} \\ \frac{Z_B-Z_A}{2} & \frac{Z_B+Z_A}{2} \end{pmatrix}$$

Igualando:

$$\begin{cases} 2\,\frac{s^2+2}{s} = \frac{Z_B+Z_A}{2} \\ 2\,\frac{s^2+1}{s} = \frac{Z_B-Z_A}{2} \end{cases}$$

$$4s + \frac{8}{s} = Z_B + Z_A$$

$$4s + \frac{4}{s} = Z_B - Z_A$$

$$Z_B = 4s + \frac{4}{s} + Z_A$$

$$\cancel{4s} + \frac{8}{s} = \cancel{4s} + \frac{4}{s} + 2Z_A$$

$$\frac{4}{s} = 2Z_A \rightarrow Z_A = \frac{2}{s}$$

$$\therefore \; Z_B = 4s + \frac{6}{s}$$

Circuito Equivalente.